

foreword

New roles for structure in biology and drug discovery

Robert B. Russell and Drake S. Eggleston



These are exciting times for those who generate and utilize macromolecular structures. Recently, we have seen the publication of 'holy grail' high resolution structures such as part of the bacterial ribosome¹⁻⁴, a G-protein coupled receptor (GPCR)⁵ and an ion channel⁶. These magnificent accomplishments show that there are no real limitations to determining three-dimensional structures of considerable size and complexity, and they stem from decades of progress in all scientific areas relating to structure determination. They also bode well for more widespread access to all structures of all types for exploring approaches to disease modulation. Indeed, the days when structures were a luxury restricted to small, easily obtainable proteins are clearly a thing of the past.

Structural genomics promises to capitalize upon numerous advances in science and technology to change our appreciation and understanding of biological systems forever. With the potential to impact heavily on the design of new pharmaceuticals, structural genomics will take a place alongside high throughput chemistry and screening as an integral platform approach underpinning modern drug discovery. In the form of a new interdisciplinary endeavor attempting to capitalize on technical advances on a grand scale, the ultimate aim of its practitioners will be to provide structural information for all known proteins. Like the large-scale genomic sequencing projects that have been running for more than a decade, this will involve profound changes in thinking and approach. Instead of developing a specific biological justification in advance of working on a protein, crystallographers and NMR spectroscopists can now consider the determination of structures for all proteins in an organism. Thus, in at least one manifestation of such an endeavor, this implies a potentially difficult move away from hypothesis driven research to a system of solving structures first and asking questions later.

At first glance, the task at hand may appear simple — escalate what has been happening in determining macromolecular structures for the last ten years — but far more is involved. An initiative of this type will benefit from coordinated efforts as have never been seen before, since expertise is required in many sub-disciplines spanning biology, chemistry and physics, and the contribution each makes must change significantly from what it has been in the past.

Protein production is a major part of any structural genomics initiative⁷. Although recombinant expression techniques are well established, generating large amounts of proteins of sufficient purity for NMR or crystallographic studies is still done on a case-by-case basis. Industrial approaches do exist for expressing large numbers of proteins to support, for example, high throughput

screening, but these must be adapted to meet the quality and quantity demands of a structural genomics approach.

Recent advances have produced an increase in the speed of macromolecular structure determination⁸⁻¹⁰. For X-ray crystallography, developments like seleno-methionine derivatives, cryo-freezing, robotic crystallization, and synchrotron radiation sources have meant that structures can be solved with smaller amounts of protein and with fewer crystals than were necessary previously. For NMR, advances in magnet and probe technology and in experimental methods such as TROSY¹¹ have expanded the range of proteins amenable to structure determination.

Bioinformatics also plays several roles in structural genomics. Target selection involves database interrogation, sequence comparison and fold recognition, to aid selection of the best candidate proteins given a particular set of requirements (for example, disease associated genes, or those that are common to most organisms)¹². Solved structures must be placed into their appropriate genomic context¹³, and annotated so that functional details may be predicted. Structural annotation may prove tricky, since large numbers of proteins of known structure but of unknown function have not been a major issue before. Comparative modeling plays an essential role by providing structures for homologs of those determined experimentally¹⁴, and efficient archiving of structural information is essential if the biological community is to make best use of all data¹⁵.

Which structures and in which order?

A major issue for proponents of structural genomics to address is that of target selection¹², or which structures from which species should be solved, and in which order? At present this is not internationally coordinated, with individual groups choosing to focus on a particular organism, such as a hyperthermophile, or a class of proteins, for their own reasons. Since it is not feasible in the short term to solve structures for all proteins from all organisms, and it is quite possible that different groups could solve the same structure, it may prove valuable to coordinate target selection to obtain reasonable coverage of protein fold or superfamily space in the shortest time possible.

Known structure, unknown function

One solved structure may reveal an unsuspected similarity to another, providing a possible evolutionary link between large protein sequence families that were previously thought not to be related^{16,17}, allowing the function of one family of proteins to be predicted from that of the other. However, a structure will not

always give insights into function, for example when a protein adopts a new fold¹⁸, or a fold that performs many functions^{19,20}. Even in the absence of fold similarities, examination of key active site residues²¹, or protein surfaces²², or, more fortuitously, the presence of a bound ligand^{6,23} can give strong clues as to function.

Structural genomics and drug discovery

It is clear that access to three-dimensional macromolecular structures makes a difference to drug discovery. Starting in the late 1980s and accelerating into the present day, insights gleaned from individual target structures have resulted in a tangible effect on the discovery of medicines which reached the market (see for example ref. 24), in addition to many more which did not survive development for reasons such as toxicity and pharmacokinetics. Both lead optimization (the process by which a small organic molecule is refined and elaborated to produce one of potency and selectivity for a target and with suitable physico-chemical properties to become a drug), and lead generation (the process of developing and screening databases of chemical entities for activities against drug targets), are greatly aided if the three-dimensional structure of the biological target, or members of a target class, are known. This is particularly so if complexes between the drug and the target can be obtained. Thus, as more protein structures become available, there will ensue an increase in the rate at which lead molecules for modulating target functions are produced and optimized, ultimately generating an increased flow of drug candidates to the clinic.

However, it is not only structures of drug targets that will aid drug discovery. Structural genomics is part of a wider functional genomics effort, and as such it promises to enhance greatly the understanding of complex biological phenomena, and to assign functions to proteins within complex biological pathways. Every pharmaceutical company is faced with the challenge of wading through a long list of new genes and investigating those that are involved in pathways of therapeutic interest. Many genes code for proteins of unknown function and structural genomics, together with other areas of functional genomics (such as gene expression, proteomics, gene knock-outs, and whole genome comparisons), will add functional understanding to genes on a large scale, both individually and collectively, as we move toward a complete biochemical and mechanistic understanding of mammalian and bacterial species.

Access to structures will also aid the design and identification of tool compounds useful for probing biological function in greater depth. Many proteins and pathways also will have implications for understanding the side effects of drug candidates. Minimizing unwanted activities is as important as enhancing desired ones in reducing lead optimization cycle times and increasing the rate of entry of drug candidates into human testing. Therefore, there is the prospect of developing structure-based knowledge of those proteins that should be avoided in drug discovery programs, to overcome complicating factors such as drug metabolism and toxicology that often negate the best efforts of skilled chemists in optimizing the activity of molecules against targets and compromise the success of clinical trials.

New antibiotics

A successful structural genomics effort could be of benefit particularly to drug discovery in the arena of antibiotic development. Although there are efforts to identify antibiotics that selectively target particular pathogenic bacteria, most drug dis-

covery efforts seek broad spectrum agents, where a single molecule is capable of acting against many pathogens. It is here where strategies such as determining the structures of proteins from thermophilic organisms, although seemingly remote from primary clinical targets such as *Escherichia coli*, *Haemophilus influenzae* and *Staphylococcus aureus*, could benefit drug discovery. This is particularly true when thermophilic proteins, such as those from *Aquifex aeolicus*²⁵, are close homologs of their pathogenic cousins. Ultimately, producing structures for targets shown to be essential for bacterial growth and survival should be the goal, and this will require and benefit from advances in capabilities for producing sufficient material from a broad spectrum of both Gram positive and Gram negative organisms.

Human disease

The situation is different for mammalian targets, which for the pharmaceutical industry generally means human proteins. There is a long history in drug discovery of pursuing classes of proteins that make the best drug targets. The usual suspects include GPCRs, ion channels, nuclear hormone receptors, proteases, kinases, integrins and DNA processing enzymes such as helicases or gyrases. Although many targets are soluble proteins amenable to a structural genomics approach, GPCRs and ion channels, which together comprise more than 50% of human drug targets currently, are integral membrane proteins that have long presented great challenges to NMR and crystallography because of problems in over-expression, crystallization and solubility. The recent high resolution structures of representatives from both of these families^{5,6} show that there are no limitations to what determined investigators can accomplish. Nevertheless, we are still a long way from high-throughput structure determination for such difficult proteins²⁶.

Academic versus industrial structural genomics

Although a structural genomics initiative will clearly benefit drug discovery, it is important to consider where academic and industrial aims are likely to differ. Addressing possible conflicts will be essential if collaborations between industrial and academic partners are to become a reality^{27,28}.

Target selection is perhaps the most obvious conflict. It will be tempting for academic efforts to gather randomly the 'low-hanging fruit', that is, proteins that can be easily expressed, particularly if success is measured by the numbers of structures produced. Although this approach should gradually lead to the solution of structures for all families of protein folds, and will provide some useful starting points for functional genomics, the numbers of actual drug targets produced could be minimal. This will be particularly true if human proteins are generally avoided, or if those that are addressed are involved primarily in protein-protein interactions, which have to date been among the most challenging and problematic in terms of inhibition by small drug-like molecules.

Another potential conflict arises from the concept that once a structure for one member of a family is solved, the academic investigator should move on to another family. This would have serious limitations for anti-bacterial research where multiple structures from the same homologous family are desirable. In the design of broad spectrum antibiotics, it is always preferable to have many structures available, preferably from many different pathogens. In this case, the rewards for discovering new antibiotics to fight the looming scourge of bacterial resistance will lie in the subtle detail of distinction. While it is possible that

foreword

comparative modelling¹⁴ may go some way towards filling the gaps, for many bacterial orthologs, low sequence similarities prevent accurate models from being constructed with current technology. A similar situation exists for human drug targets, where toxicity concerns could mean that a drug should only target one of a group of closely related proteins, and the structures of all would assist the optimization of selectivity.

A lack of emphasis placed on integral membrane proteins, which form the majority of drug targets currently, is another potential source of conflict. Proponents of structural genomics could do far worse than direct a significant effort towards such proteins. Challenges remain for the development of generic, automatable approaches for integral membrane proteins, but these appear largely to reside within the realms of gene expression, protein purification and crystallization, rather than in the solution of structures given quality X-ray diffraction or NMR data. For membrane proteins, the goals are longer-term than those based around structures that are easier to solve, but the benefits to science and to improving healthcare would be immense.

Affiliations

R.B.R. is a Senior Investigator in Bioinformatics and D.S.E. is the Director of Computational and Structural Sciences at SmithKline Beecham Pharmaceuticals.

1. Ban, N., Nissen, P., Hansen, J., Moore, P.B. & Steitz, T.A. *Science*, **289**, 905–920 (2000).
2. Schlutzen, F. *et al. Cell* **102**, 615 (2000).
3. Wimberly, B.T. *et al. Nature* **407**, 327–339 (2000).
4. Carter, A.P. *et al. Nature* **407**, 340–348 (2000).
5. Palczewski, K. *et al. Science* **289**, 739–745 (2000).
6. Doyle, D.A. *et al. Science* **280**, 69–77 (1998).
7. Edwards, A.M. *et al. Nature Struct. Biol.* **7**, 970–972 (2000).
8. Abola, E., Kuhn, P., Earnest, T. & Stevens, R.C. *Nature Struct. Biol.* **7**, 973–977 (2000).
9. Lamzin, V. & Perrakis, A. *Nature Struct. Biol.* **7**, 978–981 (2000).
10. Montelione, G.T., Zheng, D., Huang, Y.J., Gunsalus, K.C. & Szyperski, T. *Nature Struct. Biol.* **7**, 982–985 (2000).
11. Pervushin, K., Riek, R., Wider, G. & Wuthrich, K. *Proc. Natl. Acad. Sci. USA* **94**, 12366–12371 (1997).
12. Brenner, S.E. *Nature Struct. Biol.* **7**, 967–969 (2000).
13. Gerstein, M. *Nature Struct. Biol.* **7**, 960–963 (2000).
14. Sanchez, R. *et al. Nature Struct. Biol.* **7**, 986–990 (2000).
15. Berman, H.M. *et al. Nature Struct. Biol.* **7**, 957–959 (2000).
16. Minn, A.J. *et al. Nature* **385**, 353–357 (1997).
17. Artymiuk, P.J., Poirrette, A.R., Rice, D.W. & Willett, P. *Nature* **388**, 33–34 (1997).
18. Yang, F., Gustafson, K.R., Boyd, M.R. & Wlodawer, A. *Nature Struct. Biol.* **5**, 763–764 (1998).
19. Russell, R.B., Sasieni, P.D., Sternberg, M.J.E. *J. Mol. Biol.* **282**, 903–918 (1998).
20. Thornton, J.M., Orengo, C.A., Todd, A.E. & Pearl, F.M. *J. Mol. Biol.* **293**, 333–342 (1999).
21. Ho, Y.S. *et al. Nature* **385**, 89–93 (1997).
22. Boggon, T.J., Shan, W.S., Santagata, S., Myers, S.C. & Shapiro, L. *Science*, **286**, 2119–2125 (1999).
23. Zarembinski, T.I. *et al. Proc. Natl. Acad. Sci. USA* **95**, 15189–15193 (1998).
24. von Itzstein, M. *et al. Nature* **363**, 418–423 (1993).
25. Deckert, G. *Nature* **392**, 353–358 (1998).
26. Hol, W.G.J. *Nature Struct. Biol.* **7**, 964–966 (2000).
27. Butler, D. *Nature* **406**, 923–924 (2000).
28. Williamson, A.R. *Nature Struct. Biol.* **7**, 953 (2000).